EXERCISE – AI INVEST

Investment evaluation and IRR analysis with a private equity fund

A private equity fund is evaluating an investment in a new high-growth business unit operating in the software industry.

The investment takes place at time 0, and the fund plans to exit the investment at the end of year 5 (t = 5) through the sale of its equity stake.

Operating data

- Expected EBITDA at time 5:
  - EBITDA (time 5) = €18 million
- Operating cash flow to EBITDA conversion ratio:
  - k = 55%
- Long-term growth rate:
  - g = 4%
- Discount rate:
  - r = 14%
- The company has no debt, therefore:
  - Enterprise Value = Equity Value

Investment data

- Initial investment at time 0:
  - €30 million
- Holding period:
  - 5 years
- Equity stake acquired by the fund:
  - 55%
- Required target return of the fund:
  - 22% annual IRR
- No intermediate dividends or cash flows are distributed during the investment period.

Questions

1. Determine the implied EV/EBITDA multiple using the fundamentals (k, r, g).
2. Calculate the enterprise value of the company at time 5.
3. Calculate the value of the fund's equity stake at exit.
4. Calculate the IRR generated by the investment.
5. Compare the investment IRR with the target IRR required by the fund:
   - Is the investment acceptable for the fund?
   - Why?
6. Determine:

    - the post-money valuation
    - the pre-money valuation
7. Determine the Exit value required for the investment to match the target IRR.